



RV College of Engineering®
Mysore Road, RV Vidyanikethan Post,
Bengaluru - 560059, Karnataka, India

Question Paper ID: 597330

Course Code: CV375TGH

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RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

VII Semester B.E. Regular Examination February/March 2026
Freight Transportation Systems and Logistics (Institutional Elective)

Time : 3 Hours

Maximum Marks : 100

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	Mention any two issue related to intermodal freight transport.	02	1	2
1.2	Mention how commodity type influences freight travel.	02	1	2
1.3	List any two purpose of freight demand estimation.	02	2	1
1.4	List the importance of data in freight planning.	02	2	1
1.5	Explain the importance of regulation in freight transport.	02	2	2
1.6	Mention importance of route planning in freight transport.	02	2	2
1.7	Mention the importance of logistics in supply chains.	02	3	2
1.8	What is network models in logistics ?	02	3	2
1.9	List any two benefit learned from successful logistics systems.	02	4	2
1.10	Mention how AI supports logistics decision-making.	02	4	2

Part B

Question No	Question	M	CO	BT
2a	Apply freight demand modeling for a growing urban area.	08	1	3
2b	Apply freight characteristics to analyze last-mile delivery challenges.	08	1	3

3a	Analyse the key differences between passenger and freight demand models.	08 2 3
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3b	A warehouse in an industrial zone handles 450 tons of goods daily. The freight trip rate is 0.75 trips per ton for outbound delivery. Calculate: a) Total freight trips generated per day. b) If each truck has an average payload of 12 tons, find the number of truck trips required.	08 2 3
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OR

4a	Discuss freight trip rate models and their importance.	08 2 3
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4b	Discuss the major problems in freight transportation at urban and regional levels	08 2 3
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5a	Discuss the importance of scheduling in improving freight transport efficiency.	08 2 3
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5b	Explain the process of freight flow distribution from production to consumption zones.	08 2 3
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OR

6a	Explain freight supply and capacity constraints in freight transportation.	08 2 3
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6b	Describe Origin-Destination (O-D) modeling for freight transport planning.	08 2 3
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7a	Discuss methods used to adjust logistics capacity under fluctuating demand.	08 3 3
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7b	Explain the strategic importance of logistics in modern organizations.	08 3 3
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OR

8a	Describe tactical planning and scheduling in logistics operations and their significance.	08 3 3
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8b	Explain the concept of logistics and describe its major functional activities.	08 3 3
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9a	Discuss applications of big data analytics in freight and logistics decision-making.	08 4 3
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9b	Explain the role of IoT in real-time logistics monitoring.	08 4 3
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OR

10a	Evaluate the importance of distribution centers in improving freight performance. Provide examples of their strategic roles.	08 4 3
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10b	Compare various competing modes of transport for specific commodities. How does mode selection affect cost and service levels?	08 4 3
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